

Sports Nutrition & Dietetics Diploma  
Module 1

# Advanced Macronutrient Science

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## Module Overview

This module is part of the **Sports Nutrition & Dietetics Diploma** offered by Royal Fitness Club Academy. In this module, you will explore the key concepts and practical applications essential to your fitness education journey.

## Topics Covered in This Module

1. Protein turnover and synthesis rates
2. Net protein balance and leucine threshold
3. Carbohydrate storage: glycogen capacity
4. Fat oxidation at rest vs exercise
5. Metabolic flexibility
6. Respiratory quotient and substrate utilization

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## Learning Objectives

- ✓ Understand and articulate the core concepts presented in this module.
- ✓ Apply the knowledge gained to real-world fitness and coaching scenarios.
- ✓ Critically evaluate evidence and best practices in this topic area.
- ✓ Complete the module assessment with a minimum score of 70% to earn your RFC Certificate.

■ Important Note

This PDF is your study companion for Module 1. Use it alongside your online course con